

Out Station rides - without limit

The Outstation Ride feature enables passengers to book rides between cities with the option of One-Way or Round Trip travel. All outstation rides are booked 30 min later from the current time to make sure the ride is fulfilled without any drop.

Scope

- Outstation rides are facilitated without any limits.
- Ride can be either one way or round trips
- All rides are booked for 30 min from the current time, to avoid ride failures

Business Objectives

- Offer seamless long-distance travel options to passengers.
- Ensure fair earnings for drivers by considering empty return trips.
- Minimize cancellations through structured fare proposals.
- Optimize manual & auto-assignment for scheduled rides.

Feature Flow & Functionalities

Passenger App Flow

- Passengers enter the drop location, based on it we will identify the out station ride.
- Passenger choose between
 - One-Way Trip (destination city only)
 - Round Trip (return to the origin city)
- Chooses ride type:
 - Ride Now (default ride start time is current time + 30mins)
 - Ride Later (Scheduled Booking)
- If Ride Later, the passenger selects the preferred date & time for the journey.
 - One way trip - Set the start date & time
 - Round trip - selects the start time & the return date and time
 - Ride can be booked one week (8 days)prior to the start date

Instant Booking Flow

- To facilitate best vehicle for outstation rides, we are not encouraging instant booking
- All rides will be booked for Current time+30Mins
- All rides are prebooked, goes to an agent and the agent will assign the vehicle for the out station rides

Trip Assign Process

- All the booked rides will be shown in the agent portal, under outstation section
- Agent will search and manually assign a vehicle and a driver to the trip
- Agent can able to edit the trip details, if requested by the passenger over phone
- Agent can also able to cancel the trip, if requested by the passenger over phone
- 15 minutes before the scheduled time, an agent should assign a driver to the ride.
- If no manual assignment is done until 10 minutes before the ride, then an auto-assignment process selects a nearby available driver and assigns the ride to him.

Trip Start Process

- Once the agent assign to the trip to a driver, the driver will get the new ride request
- Driver accepts the ride and start to move to the pickup point
- Once the driver reaches the pickup point, he enters the current odometer reading.
- The odometer reading is then displayed to the passenger.
- The passenger confirms the reading and shares the start OTP.
- The driver enters the OTP, and the ride begins.

Trip End Process

- The trip ends when the driver Swipes on end the ride CTA
- At the drop location, the driver enters the final odometer reading.
- The odometer reading is displayed to the passenger.
- The passenger confirms the reading and shares the end OTP.
- The driver enters the OTP, and the trip ends.

Fare Calculation & Business Rules

How outstation fare is calculated

- Outstation fare is calculated based on the kilometer driven, and the total trip timing.
- We will calculate the trip fare based on the driven kilometer along with the waiting charges for the additional time spent on the trip.
- To execute these calculations we need to have the below settings in our admin portal for every zone and for every vehicle category.
 - Out station fare per km - We already have this
 - Out station return fare per km - We need to add this for each vehicle category
 - Per Km travel time - We already have this for local trips, we need one more for outstation trips. In metro cities this will vary for local and out station - Zone wise
 - Out station waiting time charge per minute - Need to add this configuration - Zone wise

Pre - Ride Fare calculation - One way

Parameters needed:

- **A:** Distance between the pickup and drop location
- **B:** Distance between drop location to urban boundary of the source city
- **C:** Out station Fare & Category lower cap fare
- **D:** Out station return fare per km
- The final fare is calculated by adding - $(A \times C) + (B \times D)$

Example 1:

- **A:** 150 kms
- **B:** 60 Kms
- **C:** Rs 13.00
- **D:** Rs 10.00

Fare Calculation is: $(150 \times 13) + (60 \times 10) = \text{Rs } 2550.00$

Post - Ride Fare calculation - One way

Parameters needed:

- **A:** Actual travel distance, we will get this from the start and end odometer reading input by the driver
- **B:** Distance between the actual drop location to urban boundary of the source city
- **C:** Travel time between pickup and drop location
- **D:** Per km Travel Seconds
- **E:** Out station waiting time charges
- **F:** Driver proposed slot price (Out station fare / category lower cap)
- **G:** Out station return fare per km

Example 1:

- **A:** 150 Kms
- **B:** 60 kms
- **C:** 225 mins (13,500 Seconds)
- **D:** 1.5 mins (90 Sec)
- **E:** Rs 1.25
- **F:** Rs 13.00
- **G:** Rs 10.00

In the above scenario, the total distance travelled is 150 kms and the time taken to travel is 13,500 seconds. That is, the time taken to travel 150 kms is exactly as we define it, no extra time or waiting time is spent on this trip.

In this case, the fare calculation will be as the below:

$$\text{Formulae} = (A \times F) + (B \times G)$$

$$\text{Ride Fare} = (150 \times 13) + (60 \times 10) = \text{Rs } 2,550.00$$

Example 2:

- **A:** 150 Kms
- **B:** 60 kms
- **C:** 300 mins (18,000 Seconds)
- **D:** 1.5 mins (90 Sec)
- **E:** Rs 1.25
- **F:** Rs 13.00
- **G:** Rs 10.00

In the above scenario, the total distance travelled is 150 kms and the time taken to travel is 18,000 seconds. In this case the time taken to travel 150 kms is 300 mins 75 mins more than our average of 225 mins. In this case we need to add the waiting time charges.

In this case, the fare calculation will be as the below:

Ride fare Formulae: $((A \times F) + (B \times G)) + ((Act. travel time - Avg. travel time) \times E)$

Final Fare: $((150 \times 13) + (60 \times 10)) + ((300 - 225) \times 1.25) = Rs 2643.75$

Pre - Ride Fare calculation - Return trip

Parameters needed:

- A: Distance between the pickup → drop & drop → Pickup
- C: Out station Fare & Category lower cap
- D: Trip Start date and time
- E: Trip end date and time
- F: Derived additional waiting time
- G: Out station waiting time charges per minute

Final fare formula: $(A \times C) + (F \times G)$

Derived additional waiting time: When a user selects a round trip, then we are asking him two additional inputs, **1. Trip start date & time. 2. Trip end date & time.** With this two inputs we can able to derive, how long the vehicle is needed for our user. Based an the source and destination we can able to derive what's the average time needed to complete the trip. The time remaining in the trip is the derived additional waiting time.

For ex: A passenger is requesting for a round trip from Point A to Point B - The total distance is 200 kms. The average time taken to complete this 200 kms is 4 Hours. If the passenger requested the vehicle for 10 hours, then additional 6 hour is the waiting time.

Example 1:

- A: 200 kms
- C: Rs 13.00
- D: 22 March 10:00 PM
- E: 23 March 6:00 AM
- F: 4 Hours
 - User requested duration - 8 Hours
 - Average time taken for 200kms - 4 Hours
 - Additional waiting time: $8 - 4 = 4$ Hours (240 minutes)
- G: Rs 2.00 per minute

Fare calculation: $(200 \times 13) + (240 \times 2) = \text{Rs. } 3080.00$

Post - Ride Fare calculation - Return trip

Parameters needed:

- A: total travel distance, we get it from the odometer reading captured at the start and end of the trip
- C: Driver Propose time selected per km fare
- D: Trip Start date and time - When driver clicked on start the trip
- E: Trip end date and time - When driver clicked on end the trip
- F: Derived additional waiting time
- G: Out station waiting time charges per minute

Final fare formula: $(A \times C) + (F \times G)$

General Business Rules

- Outstation rides must start from an operational city.
- Drivers can only accept or propose fares, not manually enter custom pricing.
- Ride start and end must be verified with odometer readings & OTP to prevent disputes.
- Scheduled rides must be assigned before 15 minutes of the ride time to avoid cancellations.